Week 5 progressive hints

Use the hints in order. After each hint, return to your own work and continue until you reach a new, specific impasse.

`W05-Q01-PA`

1. Differentiate with respect to p and w separately, keeping the other variable fixed, and place the results in a column.
2. Use $\mathrm{d}c = \nabla c(p, w)^{\mathsf{T}}[\mathrm{d}p, \mathrm{d}w]^{\mathsf{T}}$, then substitute the base point before the stated small changes.

`W05-Q01-PB`

1. Regard z as a function of the three arguments x, y and θ. The logarithm requires $y(\theta) > 0$.
2. Write $z' = z_x x' + z_y y' + z_\theta$, retaining the direct partial derivative with respect to θ.
3. Evaluate x, y, x' and y' at zero before combining the three terms.

`W05-Q02-PA`

1. The scalar implicit-function condition is $F_x \neq 0$ at the point, in addition to differentiability nearby.
2. Differentiate $F(x(\theta), \theta) = 0$ to obtain $F_x x' + F_\theta = 0$, then evaluate the two partial derivatives.

`W05-Q02-PB`

1. Form the 2×2 Jacobian of (F_1, F_2) with respect to (x, y), not with respect to all three variables.
2. Check its determinant at the point. Then use $\mathbf{J}_{\mathbf{F},\mathbf{x}}\mathbf{x}' + \mathbf{F}_\theta = \mathbf{0}$.
3. Substitute the point to obtain a 2×2 linear system for $x'(0)$ and $y'(0)$, and solve that system.

`W05-Q03-PA`

1. Replace x and y by tx and ty, with $t > 0$, and collect the resulting power of t; then compare $xf_x + yf_y$ with f.
2. For r, compare $r(tx, ty) = tx + t^2y^2$ with a possible single factor $t^k r(x, y)$.
3. At $(2, 1)$, compute $xh_x + yh_y$ and compare it with the appropriate multiple of $h(x, y)$, after first scaling h and calculating its partial derivatives.

`W05-Q04-PA`

1. Replace $\mathbf{A}$ by $(\mathbf{A} + \mathbf{A}^{\mathsf{T}})/2$.
2. Use its eigenvalues, leading principal minors, or a completed-square expression for the quadratic form to determine the sign for every non-zero vector.

`W05-Q04-PB`

1. The eigenvectors $[1, 1]^{\mathsf{T}}$ and $[1, -1]^{\mathsf{T}}$ diagonalise the matrix for every a.
2. Express the two eigenvalues as functions of a and classify the cases in which both are positive, one is zero, or their signs differ.
3. Use the positive trace to rule out both eigenvalues being non-positive.

`W05-Q05-PA`

1. For C, take two points and a weight $\lambda \in [0, 1]$. Use the convexity inequality for the square function.
2. For D, try two points with the same positive y-coordinate and opposite x-coordinates, then inspect their midpoint.

W05-Q05-PB

1. Compute the Hessian and use its determinant to test whether it is positive or negative semidefinite.
2. For a level $c > 0$, rewrite the upper contour condition $xy \geq c$ as an epigraph condition involving c/x.
3. Check the curvature of $x \mapsto c/x$ on $x > 0$, and handle levels $c \leq 0$ separately.

W05-Q06-PA

1. Sum each row to obtain p_X and each column to obtain p_Y.
2. Check independence using one joint cell against $p_X(x)p_Y(y)$; a single failure disproves independence.

W05-Q06-PB

1. For each fixed y, divide the two entries in that column by the positive column total.
2. Because X takes values zero and one, $\mathbb{E}[X \mid Y = y] = \mathbb{P}(X = 1 \mid Y = y)$.
3. Weight the three conditional means by $p_Y(y)$ and compare the result with the marginal probability $\mathbb{P}(X = 1)$.

W05-Q07-PA

1. Normalise over the triangular support using $\int_0^1 \int_0^y c \, \mathrm{d}x \, \mathrm{d}y = 1$.
2. For fixed x, the admissible y-values run from x to one. For fixed y, the admissible x-values run from zero to y.
3. Divide the joint density by $f_Y(y)$ and retain both $0 < x < y$ and $0 < y < 1$ in the conditional support.

W05-Q07-PB

1. Integrate $x f_{X|Y}(x \mid y)$ over the conditional support.
2. Use $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$ with the marginal density of Y.
3. For the direct check, integrate $x f_{X,Y}(x, y)$ over the original triangle in the same order used for normalisation.

W05-Q08-PA

1. Write the two conditional means and variances of the Bernoulli distribution as functions of Y.
2. Calculate $\mathbb{E}[\mathrm{Var}(X \mid Y)]$ and $\mathrm{Var}(\mathbb{E}[X \mid Y])$ separately before adding them.
3. For the verification, use total probability to find $\mathbb{P}(X = 1)$, then apply the Bernoulli variance formula.

W05-Q09-PA

1. Form the characteristic polynomial $\det(\mathbf{K} - \lambda \mathbf{I}_2)$.
2. Solve one homogeneous system for each eigenvalue. Two independent eigenvectors in $\mathbb{R}^2$ settle diagonalizability; place them in a matrix and use the corresponding eigenvalues on the diagonal.

W05-Q10-PA

1. Treat A, A^c as a partition and use the law of total probability for C.
2. Bayes' rule divides $\mathbb{P}(C \mid A)\mathbb{P}(A)$ by the total probability just found.

W05-Q10-PB

1. Use linearity of expectation on $4 - 2U$.
2. For variance, the additive constant disappears and the coefficient is squared.